

Hybrid Search & Retrieval POC

https://github.com/aiyshwary/Search_Engine

OVERVIEW

A production-style multilingual hybrid search engine for Indian-language documents, combining BM25 lexical retrieval to rank documents with snippet-level evidence across Tamil, Hindi, Telugu and more.

IMPACT

Built a multilingual hybrid search pipeline supporting Indian-language PDFs and DOCX files, combining BM25 and IVF-PQ for accurate document retrieval with chunk-level evidence and best-snippet selection.

KEY DETAILS

1. Ingests PDF and DOCX files with digital text extraction and Tesseract OCR fallback for scanned documents.
2. Detects document language using FastText (LID176) across Tamil, Hindi, Telugu, Kannada, Bengali and more.
3. Applies SimHash near-duplicate detection and token-aware chunking with sentence-based splitting.
4. Generates multilingual embeddings using Sentence Transformers and indexes them in FAISS IVF-PQ.
5. Hybrid BM25 and ANN retrieval with document-level score aggregation and best-snippet selection per document.
6. Embeds with paraphrase-multilingual-MiniLM-L12-v2 (L2-normalized, batch 256) and queries FAISS. The final score is $0.5 \times \max(\text{BM25}) + 0.5 \times \max(\text{ANN})$ with reciprocal-rank ANN scoring.
7. Falls back to Lingua library when FastText confidence drops below 0.60, and normalizes ISO-639-1 codes for the entire pipeline for consistent language filtering.

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8. Parallelizes ingestion via ProcessPoolExecutor with (cpu_count - 2) workers and runs BM25 and FA separate threads before union.